

Geometry Unit 9 Assessment Guide

General Information	Unit 9 revisits area of figures and extends area to lateral area, surface area, and finding area with a scale factor or a dilation. Unit 9 builds students' understanding of volume by connecting it to both cross sections and rotations. Both of these concepts foreshadow calculus topics. Through building the idea of volume as a series of cross sections, students can better understand the role of B (the area of the base) in volume formulas. The formulas for surface area and volume are provided on the first page of the assessment. The lessons allow students to build their understanding so that memorizing the formula is not the purpose. Students can solve the problems without the use of a formula. The assessment has a variety of figures so that teachers may be able to ascertain which figures students may struggle with. Teachers may also wish to determine if students do better on questions with a diagram. If students are not aware of what a figure looks like, then questions without diagrams may be difficult. Teachers may also wish to determine if students find questions with composite figures as more difficult. It is not possible to assess the totality of student knowledge of these benchmarks. The possible types of figures make this a difficult task. Teachers will need to determine whether students struggle with concepts such as area, surface area, lateral area, or volume. It may be more helpful during small group instruction to ask students for the types of figures they struggle with the most. Many students struggle with conceptualizing figures for MA.912.GR.4.2 or understanding how to visualize the cross section. Manipulatives may help support these students.
Calculator Information	The questions on the unit assessment are calculator.
Total Recommended Points	The total recommended points in this guide are 100 points.

Question	1	Benchmark(s)	MA.912.GR.4.6
	241.029 ft²	Recommended Scoring (23 Total Points)	Consider distributing points as follows: • correctly determines the radius and connects it to the height of the prism (4 points) • area of the front/back of the prism (3 points) • area of the right/left of the prism (3 points) • area of the base of the prism (3 points) • area of the base of the cylinder (as one circle or as two semicircles) (3 points) • area of the lateral area of the cylinder (3 points) • total surface area (3 points) • correct units (1 point)
		Additional Notes	Students may not recognize that the height of the prism is 3 feet, which is the length of the radius. If students miss this detail the teacher is encouraged to consider this as a detail that makes this question both more difficult and have a

			higher cognitive demand. Therefore, teachers are encouraged to follow student work to determine if this is the only error a student makes. And if that is the case then the point deduction should reflect all that a student did get correct.
Question	2	Benchmark(s)	MA.912.GR.4.5
The volume of the stack of hexagonal blocks is ☐ equal to ☒ less than ☐ greater than the volume of the stack of square blocks. This is because ☐ both blocks have the same thickness. ☐ both stacks have the same number of blocks. ☐ the base area of the hexagon block is greater than the base area of the square block. ☒ the base area of the square block is greater than the base area of the hexagon block.		Recommended Scoring (12 Total Points)	Consider giving students 3 points for correct work for each: • determines the volume of the stack of hexagonal blocks • determines the volume of the stack of square blocks • correct selection in the first box • correct selection in the second box

Question	3	Benchmark(s)	MA.912.GR.4.2
	When the rectangle is rotated about the dotted line, the resulting figure is a ☐ cone ☒ cylinder ☐ square pyramid ☐ rectangular prism with a height of ☒ 2 in. ☐ 4 in. ☐ 8 in. and a ☐ radius of 2 in. ☒ radius of 4 in. ☐ radius of 8 in. ☐ length of 2 in. ☐ length of 4 in. ☐ length of 8 in.	Recommended Scoring (9 Total Points)	Consider giving students 3 points for each correct selection in each box.
Question	4	Benchmark(s)	MA.912.GR.4.6
	800 cm²	Recommended Scoring (13 Total Points)	Consider giving students 4 points for correct work for the area of each side of the prism and 1 point for the correct units.
Question	5a	Benchmark(s)	MA.912.GR.4.3, MA.912.GR.4.4
	776 mi²	Recommended Scoring (10 Total Points)	Consider distributing points as follows: - determines the area of WGRH (2 points) - determines the area of RHBX (2 points) - correctly applies the scale either before finding the area or after finding the area (3 points) - total area (2 points) - correct units (1 point) Note: Students may divide the area up differently. Teachers may need to consider how to redistribute points.

Question	5b	Benchmark(s)	MA.912.GR.4.3
	○ decrease by a factor of $\frac{1}{4}$. ○ decrease by a factor of 4. ○ increase by a factor of $\frac{1}{4}$. ○ increase by a factor of 4.	Recommended Scoring (3 Total Points)	No partial credit is suggested.
Question	5c	Benchmark(s)	MA.912.GR.4.4
	9.2 lobsters/mi ²	Recommended Scoring (7 Total Points)	Consider distributing points as follows: • determines the area of WGRH (3 points) • determines the density (3 points) • correct units (1 point)
Question	6	Benchmark(s)	MA.912.GR.4.5
	5989.5 π cm ³	Recommended Scoring (4 Total Points)	Consider distributing points as follows: • determines the volume (3 points) • correct units (1 point)

Question	7	Benchmark(s)	MA.912.GR.4.5
	2769.592 cm ³	Recommended Scoring (10 Total Points)	Consider distributing points as follows: • determines the volume of the prism (3 points) • determines the volume pyramid (3 points) • determines the total volume (3 points) • correct units (1 point)
Question	8	Benchmark(s)	MA.912.GR.4.1
☐ A plane parallel to the base of a cone ☐ A plane perpendicular to the base of a cylinder ☐ A plane parallel to the base of a triangular prism ☐ A plane parallel to the base of a rectangular prism ☐ A plane perpendicular to the base of a triangular prism ☐ A plane perpendicular to the base of a rectangular pyramid		Recommended Scoring (9 Total Points)	Consider giving 3 points for each correct choice and deducting 3 points for each incorrect choice.